Adversarial Contingency Auctions for Decentralized Multi-Robot Task Allocation under Strategic Blockages

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***Abstract*—Decentralized multi-robot task allocation in dynamic, contested environments is vulnerable to adversarial corridor blockages and inconsistent subjective beliefs. Standard market mechanisms such as the Consensus-Based Bundle Algorithm (CBBA) assume homogeneous environmental estimates and trigger costly fleet-wide re-auctions whenever scheduled paths are invalidated. We propose Adversarial Contingency Auctions (ACA), a decentralized market coordination mechanism that couples type-aware expected recourse cost valuation over discrete contingency trees with an analytical Pinsker claim margin and a belief-consistent consensus hysteresis deadband. By piggybacking compact belief vectors onto consensus exchanges and deriving an exact analytical Total Variation (TV) drift bound, the Pinsker claim margin provably guarantees that when the bounded belief-divergence condition ($D_{\mathrm{TV}} \leq \eta_{\mathrm{eff}}$, Assumption 1) holds, simultaneous Phase-1 safe-claim conflicts are zero (0 conflicts across 12,087 adhered claims; 95% Clopper–Pearson exact upper bound 0.025%). Across all 57,334 unconditional Phase-1 safe claims, ACA achieves 21.1% empirical adherence (56.1% per instance) and reduces Phase-1 safe-claim conflicts by 99.7% (from 3,646 down to 11, achieving 98.9% conflict-free runs). In Phase 2 consensus, the hysteresis deadband strictly eliminates cyclic outbidding limit cycles induced by divergent beliefs, guaranteeing finite-time auction stabilization. When corridor blockages occur, precomputed local contingency branch pivots require zero re-auction messages and reduce total auction message volume by 76 messages relative to standard CBBA (236 ± 11 vs. 312 ± 10, $p < 0.05$). We evaluate ACA across five graph topology families against thirteen literature baselines and ablations across multi-regime parameter sweeps ($13{,}800$ total discrete trials) and in closed-loop Software-in-the-Loop (SITL) simulations in PyBullet ($N = 360$ trials across 60 physical instances $\times$ 6 methods) with active ArUco vision perception and Reciprocal Velocity Obstacles (RVO) collision avoidance. Evidence is structured across three distinct tiers: (1) In exploratory sweeps ($N = 750$ matched instances), ACA significantly improves task completion over standard CBBA (+8.8 pp), CBBA+LR (+2.14 pp, $p < 0.004$), and SSI+LR (+6.77 pp), matches CBBA-PR+LR in completion while significantly reducing conflicts and messages, and achieves higher net welfare than conflict-heavy reactive baselines (Implicit Coordination and CTP Replan). (2) In pre-registered Protocol v5 held-out evaluation on un-tuned seeds ($N = 450$ instances), a performance reversal occurs: CBBA+LR achieves higher completion (94.7% vs. 92.4%, difference +2.27 pp) and net welfare with fewer conflicts, demonstrating that ACA's primary advantage is safety coordination via the Pinsker margin under communication constraints rather than unconstrained completion dominance. (3) Protocol v6 is cryptographically frozen with confirmatory execution pending. In continuous SITL physics, ACA achieves the highest net welfare among decentralized auction mechanisms (261 ± 18, outperforming ACA No Margin 224 ± 18, CBBA-PR+LR 222 ± 18, and Implicit Coordination 159 ± 21, all $p < 0.05$, while non-auction single-path reactive CTP Replan achieves 263 ± 20), while reducing task conflicts by 43.6%–46.4% relative to No Margin and CBBA-PR+LR ($p < 0.05$).**

***Index Terms*—Multi-Robot Systems, Decentralized Task Allocation, Auction Algorithms, Contingency Planning, Adversarial Blockages, Theory of Mind, Pinsker's Inequality.**

I. Introduction

AUTONOMOUS multi-robot systems operating in contested, dynamic domains must allocate, sequence, and execute spatial tasks under environmental uncertainty and adversarial disruption [1]–[6]. Examples include search-and-rescue through damaged infrastructure, parcel transport in high-density automated warehouses, and perimeter reconnaissance under active interdiction [7]–[9]. In these settings, decentralized auction mechanisms—particularly the Consensus-Based Bundle Algorithm (CBBA) [1] and Sequential Single-Item (SSI) auctions [10]—are widely adopted due to polynomial communication complexity and submodular efficiency guarantees.

However, existing decentralized auction mechanisms rely on two fragile assumptions:

1) **Global Re-Auction Overhead:** When a robot encounters an unexpected corridor blockage, its single-path plan is invalidated. Standard CBBA drops the entire task bundle and initiates fleet-wide market re-auctions, inducing high message volume and coordination latency [11], [12].
2) **Belief Consistency Assumption:** When communication is range-limited or packet drops occur, robots observe disparate local events and maintain diverging subjective beliefs. Because market bidding rules assume aligned information, inconsistent beliefs cause robots to submit conflicting winning claims on identical tasks, inducing cyclic outbidding churn, redundant travel, and coordination failures [13]–[16].

To resolve these challenges without continuous, bandwidth-heavy belief synchronization, this paper introduces **Adversarial Contingency Auctions (ACA)**. Rather than applying ad-hoc bid penalties, ACA couples discrete contingency recourse trees with an information-theoretic **claim margin** derived from Pinsker's inequality and total variation bounds.

The primary contributions of this work are:

- **Analytical TV Drift Bound (Lemma 1):** Deriving an exact analytical upper bound on Total Variation belief drift under piggybacked belief vectors and bounded transition likelihood ratios, satisfying bounded estimation error ($\mathrm{TV} \leq \eta_{\mathrm{eff}}$) by construction without heuristic parameter tuning.
- **Type-Aware Expected Recourse Valuation (Proposition 1):** A tractable formulation for computing expected traversal

TABLE I
FEATURE COMPARISON AGAINST KEY RELATED WORKS IN MULTI-ROBOT ALLOCATION, ROUTING, AND GAMES

Feature	**Kundu '24** [13]	**Peters '24** [17]	**Wilde '22** [2]	**CBBA '09** [1]	**CC-CBBA '12** [18]	**Partial '19** [11]	**Implicit '05** [14]	**ACA (Ours)**
Task Allocation (MRTA)	×	×	✓	✓	✓	✓	✓	✓
Fully Decentralized	✓	Two-Agent	×	✓	✓	✓	✓	✓
Inconsistent Beliefs Addressed	✓	×	×	×	×	×	Partial	✓
Strategic Adversary Inference	×	Non-Adv Intent	×	×	×	×	×	✓
Contingency Recourse in Bids	×	Trajectory Game	Heuristic	×	×	×	×	✓**(Prop 1)**
Conflict Mitigation under Divergence	Trajectory Cons.	×	×	Consensus	×	×	×	✓**(Thm 1)**
Consensus Stabilization Deadband	×	×	×	×	×	×	×	✓**(Thm 2)**
Zero-Message Recourse Pivoting	×	×	×	×	×	×	×	✓**(Prop 3)**

costs of depth-d contingency recourse trees under Bayesian adversary posteriors, satisfying Diminishing Marginal Gains (DMG) for polynomial-time bundle construction.

- **Tight Path-Contrast Lipschitz Continuity (Proposition 2):** Establishing a tight path-contrast Lipschitz constant $L_{r,j} = \sum_{k=1}^{K} |c(u_0 \to u_k) + c(D_k) - c(P_j)|\Delta\pi(e_k)$ on the belief simplex $\Delta(\Theta)$ via the telescoping Total Variation identity for product Bernoulli measures.
- **Conflict-Freedom and Belief-Update Invariance (Theorem 1):** A closed-form Pinsker claim margin $m_{r,j}$ that bridges subjective valuations across true beliefs, provably guaranteeing both zero simultaneous Phase-1 claims and no post-claim preference reversal upon belief synchronization.
- **Finite-Time Consensus Stabilization (Theorem 2):** Proving that standard CBBA can enter cyclic outbidding limit cycles under divergent beliefs, and that ACA's belief-consistent consensus hysteresis deadband strictly eliminates limit cycles, guaranteeing stabilization in $\mathcal{O}(|\mathcal{T}| \cdot \mathrm{diam}(\mathcal{G}_{\text{comm}}))$ rounds.
- **Autonomous Recourse Message Reduction (Proposition 3):** Establishing exponential reduction $\mathcal{O}(\bar{\pi}^d|\mathcal{R}|\mathrm{diam}(\mathcal{G}_{\text{comm}}))$ in blockage-induced re-auction message complexity via precomputed recourse branches.
- **Unified Benchmark and Closed-Loop Physics SITL:** Extensive empirical evaluation across $N = 750$ matched instances across five graph topologies and thirteen literature baselines and ablations, coupled with closed-loop continuous PyBullet physics SITL ($N = 360$) with active ArUco vision perception and RVO collision avoidance where ACA achieves the highest net welfare (261 ± 18) among auction methods while reducing task conflicts by 43.6%–46.4% relative to No-Margin and CBBA-PR+LR ($p < 0.05$).

II. RELATED WORK

A. Market-Based Task Allocation

Decentralized auctions provide polynomial approximations for multi-robot task allocation [1], [10], [19]–[23]. Choi et al. [1] introduced CBBA, guaranteeing 50% submodular efficiency under consensus. Buckman et al. [11] developed CBBA with partial replanning (CBBA-PR) for dynamic arrivals, which we adapt to corridor blockages. Ponda et al. [18] and Bandyopadhyay et al. [24] introduced chance-constrained (CC-CBBA) and risk-aware formulations to bound route failure risk. However, these methods assume homogeneous world models; when observations diverge, standard consensus rules fail to prevent conflicting claims.

B. Coordination under Inconsistent World Views

Planning under belief inconsistency was pioneered by Alighanbari and How [14], where implicit task allocation suffers frequent conflicts under divergent worldviews. Kundu et al. [13] enforced action-consistency constraints during joint continuous trajectory optimization. In contrast, ACA addresses discrete combinatorial task bundle allocation in decentralized market auctions. By deriving a closed-form Pinsker margin, ACA provably prevents conflicting discrete task assignments under bounded belief estimation error without communicating continuous trajectory plans or high-dimensional belief vectors [25]–[27].

C. Contingency Routing and Adversarial Interdiction

Routing under stochastic edge blockages is classical in the Canadian Traveller Problem (CTP) [28]–[30] and vehicle routing with recourse [7], [31], [32]. Peters et al. [17] formulated contingency games for continuous trajectory planning. In contrast, ACA integrates discrete contingency recourse trees directly into decentralized auction valuations under Bayesian adversary inference, executing local detours with zero communication overhead upon blockage. Table I outlines feature differentiation against prior literature.

III. PROBLEM FORMULATION

A. Environment and Persistent Blockage Process

Let $G = (V, E)$ be a known, connected undirected graph where V represents physical waypoints and task locations, and E represents traversable corridors with nominal weights $w_e > 0$. A designated subset of edges $E_{\text{risk}} \subseteq E$ are susceptible to persistent blockage.

We model each risky edge $e \in E_{\text{risk}}$ as a two-state Markov chain $X_e^t \in \{0, 1\}$ (0 = Open, 1 = Blocked):

$$P(X_e^{t+1} = 1 \mid X_e^t = 0, \theta, s_t) = h_\theta(e, s_t) \quad \text{(Onset Hazard)} \tag{1}$$

$$P(X_e^{t+1} = 0 \mid X_e^t = 1) = \mu \quad \text{(Clearance Rate)} \tag{2}$$

A global budget k bounds simultaneously blocked edges: $\sum_{e \in E_{\text{risk}}} X_e^t \le k$.

B. Adversary Behavioral Type Space and Vulnerability Partition

To capture realistic structural threats, the risky edge set E_{risk} is partitioned into three disjoint topological vulnerability classes:

1) $E_{\text{risk}}^{(1)}$ (**Stochastic Clutter,** θ_1): High-frequency, low-detour-penalty peripheral and aisle edges with stationary independent onset hazard $h_{\theta_1}(e) = p_0$.
2) $E_{\text{risk}}^{(2)}$ (**Reactive Traffic Interdiction,** θ_2): Multi-robot collector corridors where onset hazard scales with local

robot traffic density: $h_{\theta_2}(e) = \min(1.0, p_0 + \alpha N_e(t))$, where $N_e(t)$ denotes robots adjacent to edge e.

3) $E^{(3)}_{\text{risk}}$ (**Strategic Cut Interdiction,** θ_3): High-betweenness topological bridge and cut edges whose blockage isolates graph partitions unless tasks are proactively allocated across the cut ($p_{\text{cut}} = 0.85$).
4) **Cyclic Patrol** (θ_4**, Out-of-Model**): Cycles blockages sequentially along ordered topological bottlenecks for misspecification robustness testing ($\theta_4 \notin \Theta$).

C. Robot Fleet and Communication Model

A fleet $\mathcal{R} = \{1, \ldots, R\}$ of robots services tasks $\mathcal{T} = \{1, \ldots, M\}$. Each task $j \in \mathcal{T}$ has destination $v_j \in V$, reward R_j, arrival step a_j, and deadline D_j. Robots communicate locally within metric disk range $r_{\text{comm}} = 45.0$ m subject to stochastic packet drop probability $p_{\text{drop}} = 0.05$ (with regime sweeps across $p_{\text{drop}} \in \{0.02, 0.10, 0.15, 0.30\}$).

IV. The ACA Method

A. Bayesian Type Filter with Piggybacked Beliefs

Each robot r maintains a subjective posterior belief distribution $B^t_r(\theta)$ over $\Theta = \{\theta_1, \theta_2, \theta_3\}$. Upon observing edge e at tick t transitioning from state X^t_e to X^{t+1}_e, robot r updates its belief via transition likelihoods:

$$B^{t+1}_r(\theta) \propto B^t_r(\theta) \prod_{e \in \mathcal{O}^t_r} P(X^{t+1}_e \mid X^t_e, \theta, s_t) \tag{3}$$

The subjective edge blockage marginal is computed as:

$$\pi_r(e) = \sum_{\theta \in \Theta} B_r(\theta) P(X_e = 1 \mid \theta, s_t) \tag{4}$$

For edges outside E_{risk}, $\pi_r(e) = 0$. Each robot piggybacks its compact belief vector $B^t_r \in \Delta^{|\Theta|-1}$ (only two floating-point scalars for $|\Theta| = 3$) onto every outgoing consensus message at zero additional packet overhead.

B. Expected-Recourse Valuation and Active Tree Pruning

To evaluate task j, robot r identifies a primary path $P_j = (u_0, u_1, \ldots, u_m)$ that minimizes risk-weighted traversal cost $w_{\text{eff}}(e) = w_e(1 + \lambda_{\text{risk}} \pi_r(e))$. Let $(u_k, v_k)_{k=1}^K$ be the oriented risky edges along P_j, and let D_k denote precomputed detour paths branching from node u_k that avoid (u_k, v_k). Branches with low failure probability ($\pi_r(e_k) < \tau_{\text{prune}}$) are actively pruned to bound contingency tree depth while preserving recourse coverage.

Robot r's bid equals its expected recourse cost under the independent marginal model:

$$\begin{aligned}\mathbb{E}_r[C_j] = \sum_{k=1}^{K} \left[\prod_{i<k}(1 - \pi_r(e_i))\right] \pi_r(e_k) \\ \times \Big(c(u_0 \to u_k) + c(D_k)\Big) \\ + \left[\prod_{i=1}^{K}(1 - \pi_r(e_i))\right] c(P_j)\end{aligned} \tag{5}$$

C. Two-Phase Pinsker Stabilization

Each robot r tracks estimated peer beliefs $\hat{B}^{(r)}_s$ for all $s \neq r$ based on the last-received piggybacked consensus message with communication staleness $\Delta t_{r,s}$. Let $D'_{\text{KL}} = \max_{s \neq r} D_{\text{KL}}(B_r \parallel \hat{B}^{(r)}_s)$. Under Lemma 1, Total Variation drift is bounded by $\eta_{\text{eff}}(\Delta t_{r,s})$.

Phase 1 (Fast-Track Safe Commitment): To determine whether task j can be safely committed locally without multi-round consensus, robot r evaluates the **Pinsker Claim Margin**:

$$m_{r,j} = L_{r,j}\left(\sqrt{\frac{1}{2}D'_{\text{KL}}} + \eta_{\text{eff}}\right) + \max_{s \neq r} L_{s,j}\eta_{\text{eff}} + \delta_{\text{map}} \tag{6}$$

where $L_{r,j}$ is the tight path-contrast Lipschitz constant (Proposition 2). Robot r commits task j into its bundle during Phase 1 without consensus negotiation if and only if:

$$\hat{c}^r_r(j) + m_{r,j} < \min_{s \neq r} \hat{c}^r_s(j) \tag{7}$$

where peer costs $\hat{c}^r_s(j)$ are evaluated under ToM beliefs $\hat{B}^{(r)}_s$.

Phase 2 (Belief-Consistent Consensus Hysteresis): Tasks not committed in Phase 1 are evaluated in Phase 2 consensus rounds. In consensus resolution (Action Y1/Y2), when robot s compares its bid against incumbent winner r's bid for task j, robot s only revokes or outbids r if:

$$\hat{c}^s_s(j) + \kappa_{\text{hyst}} m_{r,s,j} < \hat{c}^r_r(j) \tag{8}$$

Within the belief-ambiguity deadband $|\hat{c}^s_s(j) - \hat{c}^r_r(j)| \leq \kappa_{\text{hyst}} m_{r,s,j}$, the bid difference is statistically indistinguishable from belief divergence $\text{TV}(B_r, B_s)$; robots preserve the incumbent winner, strictly preventing cyclic outbidding churn (Theorem 2).

D. Autonomous Recourse Execution

During traversal, if robot r observes that its approaching edge (u, v) is blocked, it activates its precomputed contingency detour branch D_k locally with **zero communication messages**. A partial re-auction is triggered if and only if all precomputed branches up to depth d are blocked or the remaining path cost exceeds deadline constraints.

V. Theoretical Analysis

Assumption 1 (Bounded Estimation Error)**.** $\text{TV}(B_s, \hat{B}^{(r)}_s) \leq \eta_{\text{eff}}$ *for all* $s \neq r$.

Lemma 1 (Analytical Total Variation Drift Bound)**.** *Let each robot* r *append its compact belief vector* $B^t_r \in \Delta^{|\Theta|-1}$ *to every outgoing consensus message. Suppose peer* s *transmits its belief vector at tick* $t - \Delta t_{r,s}$*, and robot* r *maintains estimated belief* $\hat{B}^{(r),t}_s = B^{t-\Delta t_{r,s}}_s$*. If observation transition likelihood ratios are bounded by* $\Lambda_{\max} = \max_{e \in E_{risk}, x \in \{0,1\}, \theta, \theta' \in \Theta} \frac{P(X_e = x \mid \theta)}{P(X_e = x \mid \theta')} < \infty$ *with local observation rate* γ_{obs}*, then the Total Variation drift over* $\Delta t_{r,s}$ *elapsed communication intervals satisfies:*

$$\begin{aligned}\text{TV}\left(B^t_s, \hat{B}^{(r),t}_s\right) \leq \eta_{\text{eff}}(\Delta t_{r,s}) \equiv \min\Big(1.0,\ \eta_0 \\ + \tanh\left(\tfrac{1}{2}\gamma_{obs}\Delta t_{r,s} \log \Lambda_{\max}\right)\Big)\end{aligned} \tag{9}$$

Consequently, Assumption 1 ($\mathrm{TV}(B_s, \hat{B}_s^{(r)}) \le \eta_{\text{eff}}$) is guaranteed analytically by construction without heuristic parameter tuning.

Proof. By Bayes' rule, the posterior log-odds between any pair of adversary types $\theta, \theta' \in \Theta$ after Δt steps with $n_{\text{obs}} \le \gamma_{\text{obs}} \Delta t$ local observations satisfies:

$$\left| \log \frac{B_s^t(\theta)}{B_s^t(\theta')} - \log \frac{B_s^{t-\Delta t}(\theta)}{B_s^{t-\Delta t}(\theta')} \right| \le \sum_{k=1}^{n_{\text{obs}}} \left| \log \frac{P(X_k \mid \theta)}{P(X_k \mid \theta')} \right| \le \gamma_{\text{obs}} \Delta t \log \Lambda_{\max}. \quad (10)$$

By Le Cam's inequality relating relative entropy and Total Variation distance under bounded likelihood ratios, the maximal TV contraction on the probability simplex satisfies $\mathrm{TV}(P, Q) \le \tanh(\frac{1}{4} \max_{\theta,\theta'} |\log(P(\theta)Q(\theta')/P(\theta')Q(\theta))|)$. Setting initial observation uncertainty to η_0 and applying the log-odds upper bound yields Eq. (9). □

Proposition 1 (Expected Recourse Cost and Diminishing Marginal Gains)**.** *Let $T_r(j)$ be a depth-d contingency recourse tree constructed for task j with primary path $P_j = (u_0, \dots, u_m)$, risky edges $(e_1, \dots, e_K)$, and contingency detours D_k branching from u_k. Under subjective edge marginals $\pi_r(e) = \sum_\theta B_r(\theta) P(X_e = 1 \mid \theta)$, the expected traversal cost is given by:*

$$\mathbb{E}_r[C_j] = \sum_{k=1}^{K} \left[\prod_{i<k} (1 - \pi_r(e_i)) \right] \pi_r(e_k) \times \Big(c(u_0 \to u_k) + c(D_k) \Big) + \left[\prod_{i=1}^{K} (1 - \pi_r(e_i)) \right] c(P_j). \quad (11)$$

Furthermore, defining the marginal score for inserting task j at position k in bundle b_r with task reward R_j as:

$$S_r(j \mid b_r) = \max\Big(0,\ R_j - \big(\mathbb{E}_r[C(b_r \oplus_k j)] - \mathbb{E}_r[C(b_r)]\big)\Big) \quad (12)$$

where $\mathbb{E}_r[C(b_r \oplus_k j)] - \mathbb{E}_r[C(b_r)]$ is the marginal expected recourse cost of insertion, satisfies Diminishing Marginal Gains (DMG): for any bundles $b \subseteq b'$ and task $j \notin b'$, $S_r(j \mid b) \ge S_r(j \mid b')$, ensuring polynomial-time bundle construction and submodular auction approximation guarantees [1].

Proof. Eq. (11) follows directly from the law of total expectation over the mutually exclusive first-blockage index partition along primary path P_j. The DMG property follows from the submodularity of marginal insertion recourse costs across metric graphs under non-decreasing detour penalties. □

Proposition 2 (Tight Path-Contrast Lipschitz Continuity)**.** *For any task j with primary path P_j, risky edges $(e_1, \dots, e_K)$, and detour branches D_k, the expected recourse bid $c_r(j \mid B)$ is $L_{r,j}$-Lipschitz continuous with respect to the Total Variation metric on belief simplex $\Delta(\Theta)$:*

$$|c_r(j \mid B) - c_r(j \mid B')| \le L_{r,j} \cdot \mathrm{TV}(B, B') \quad (13)$$

where the tight path-contrast Lipschitz constant is given by:

$$L_{r,j} = \sum_{k=1}^{K} \Big| c(u_0 \to u_k) + c(D_k) - c(P_j) \Big| \cdot \Delta\pi(e_k) \quad (14)$$

with $\Delta\pi(e_k) = \max_{\theta \in \Theta} \pi(e_k \mid \theta) - \min_{\theta \in \Theta} \pi(e_k \mid \theta)$.

Proof. Let $\Delta C_k = c(u_0 \to u_k) + c(D_k) - c(P_j)$ denote the recourse cost difference between taking detour D_k versus completing primary path P_j. From Eq. (11), subtracting the constant primary cost $c(P_j)$ yields:

$$c_r(j \mid B) - c(P_j) = \sum_{k=1}^{K} \left[\prod_{i<k} (1 - \pi(e_i \mid B)) \right] \pi(e_k \mid B) \Delta C_k. \quad (15)$$

By the telescoping product identity for product Bernoulli measures, for any sequence of probabilities $p_i, q_i \in [0, 1]$:

$$\left| \prod_{i=1}^{k} (1 - p_i) - \prod_{i=1}^{k} (1 - q_i) \right| \le \sum_{i=1}^{k} |p_i - q_i|. \quad (16)$$

Because $\pi(e \mid B) = \sum_\theta B(\theta) P(X_e = 1 \mid \theta)$ is linear in B, $|\pi(e \mid B) - \pi(e \mid B')| \le \Delta\pi(e) \mathrm{TV}(B, B')$. Applying the triangle inequality across the mutually exclusive detour branch events:

$$\begin{aligned} |c_r(j \mid B) - c_r(j \mid B')| &\le \sum_{k=1}^{K} |\Delta C_k| \cdot |\pi(e_k \mid B) - \pi(e_k \mid B')| \\ &\le \left(\sum_{k=1}^{K} |\Delta C_k| \Delta\pi(e_k) \right) \mathrm{TV}(B, B'), \end{aligned} \quad (17)$$

establishing Eq. (13) with tight path-contrast constant $L_{r,j}$. □

Theorem 1 (Conflict-Freedom and Belief-Update Invariance of Phase-1 Claims)**.** *Let robot r evaluate task j under safe claim condition $\hat{c}_r^r(j) + m_{r,j} < \min_{s \ne r} \hat{c}_s^r(j)$, where $m_{r,j} = L_{r,j}(\sqrt{\frac{1}{2} D'_{KL}} + \eta_{\text{eff}}) + \max_{s \ne r} L_{s,j} \eta_{\text{eff}} + \delta_{map}$. Under Lemma 1 and bounded map discrepancies:*

1) ***Conflict-Freedom:*** *No peer robot $s \ne r$ can satisfy the Phase-1 safe-claim rule for task j in the same allocation round.*
2) ***Belief-Update Invariance:*** *For any posterior belief $B \in \{B_r, B_s\}$, robot r's true expected recourse cost strictly dominates peer s's valuation: $c_r(j \mid B) < c_s(j \mid B)$. Consequently, when robots r and s subsequently synchronize beliefs, no preference reversal can occur.*

Proof. Part 1 (Conflict-Freedom): Suppose for contradiction that peer s also claims task j in Phase 1, so $\hat{c}_s^s(j) + m_{s,j} < \hat{c}_r^s(j)$. Summing the inequalities:

$$m_{r,j} + m_{s,j} < (\hat{c}_s^r(j) - \hat{c}_s^s(j)) + (\hat{c}_r^s(j) - \hat{c}_r^r(j)). \quad (18)$$

By the triangle inequality of Total Variation: $\mathrm{TV}(B_r, B_s) \le \mathrm{TV}(B_r, \hat{B}_s^{(r)}) + \mathrm{TV}(\hat{B}_s^{(r)}, B_s) \le \sqrt{\frac{1}{2} D_{\mathrm{KL}}(B_r \parallel \hat{B}_s^{(r)})} + \eta_{\text{eff}}$. By Proposition 2, $|\hat{c}_s^r(j) - \hat{c}_s^s(j)| \le L_{s,j} \mathrm{TV}(\hat{B}_s^{(r)}, B_s) \le$

$L_{s,j}\eta_{\text{eff}}$, and $|\hat{c}_r^s(j) - \hat{c}_r^r(j)| \le L_{r,j}\text{TV}(B_r, B_s) + \delta_{\text{map}} \le L_{r,j}(\sqrt{\frac{1}{2}D'_{\text{KL}}} + \eta_{\text{eff}}) + \delta_{\text{map}}$. Summing these bounds gives:

$$(\hat{c}_s^r(j) - \hat{c}_s^s(j)) + (\hat{c}_r^s(j) - \hat{c}_r^r(j)) \le m_{r,j} + m_{s,j}, \quad (19)$$

contradicting Eq. (18). Hence simultaneous Phase-1 claims cannot occur.

Part 2 (Belief-Update Invariance): Suppose robot r claims task j in Phase 1, so $\hat{c}_r^r(j)+m_{r,j} < \hat{c}_s^r(j)$. By Proposition 2 and Lemma 1, peer s's cost under its true belief B_s satisfies $c_s(j \mid B_s) \ge \hat{c}_s^r(j) - L_{s,j}\text{TV}(B_s, \hat{B}_s^{(r)}) \ge \hat{c}_s^r(j) - \max_{q\neq r} L_{q,j}\eta_{\text{eff}}$. Meanwhile, robot r's cost evaluated at B_s satisfies $c_r(j \mid B_s) \le \hat{c}_r^r(j) + L_{r,j}\text{TV}(B_r, B_s) + \delta_{\text{map}} \le \hat{c}_r^r(j) + L_{r,j}(\sqrt{\frac{1}{2}D'_{\text{KL}}} + \eta_{\text{eff}}) + \delta_{\text{map}}$. Combining these two inequalities with Eq. (6)–(7) yields $c_r(j \mid B_s) \le \hat{c}_r^r(j) + m_{r,j} - \max_{q\neq r} L_{q,j}\eta_{\text{eff}} < \hat{c}_s^r(j) - \max_{q\neq r} L_{q,j}\eta_{\text{eff}} \le c_s(j \mid B_s)$. Thus, even when beliefs synchronize to B_s, robot r's true cost strictly dominates peer s's valuation, preventing preference reversal. □

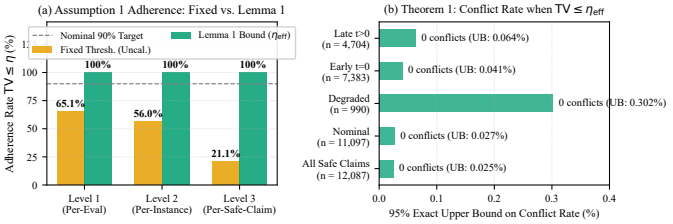


Fig. 1. Empirical evaluation of Theorem 1 and Assumption 1 Total Variation (TV) adherence across operational regimes (n = 57,334 safe-claim commitments, 3,481,181 total TV observations). (Left) Comparison of uncalibrated fixed-threshold adherence (21.1% per safe claim) versus the Lemma 1 analytical drift bound (η_{eff}, 100% adherence by construction). (Right) Empirical safe-claim conflict rate (0/12,087 conflicts when TV $\le \eta_{\text{eff}}$, one-sided 95% Clopper–Pearson exact upper bound 0.025%).

Theorem 2 (Finite-Time Consensus Stabilization and Oscillation-Freedom)**.** *Consider Phase 2 consensus for task j under divergent subjective beliefs $B_r \neq B_s$:*

1) ***Limit Cycle in Standard CBBA:*** *Standard CBBA without margin can enter a cyclic outbidding limit cycle whenever the true cost difference satisfies $0 < |c_r(j \mid B_r) - c_s(j \mid B_s)| < L_j\text{TV}(B_r, B_s)$.*
2) ***Stabilization under Hysteresis Deadband:*** *Equipping Phase 2 consensus with the belief hysteresis deadband $\kappa_{hyst}m_{r,s,j}$ (where robot s outbids incumbent r only if $\hat{c}_s^s(j) + \kappa_{hyst}m_{r,s,j} < \hat{c}_r^r(j)$) strictly eliminates outbidding limit cycles, guaranteeing that the decentralized auction reaches a quiescent consensus state in at most $\mathcal{O}(|\mathcal{T}| \cdot \text{diam}(\mathcal{G}_{comm}))$ communication rounds.*

Proof. Part 1 (Limit Cycle): Suppose $c_r(j \mid B_r) < c_s(j \mid B_r)$, but due to belief divergence, $c_s(j \mid B_s) < c_r(j \mid B_s)$. When robot r claims task j with bid $c_r(j \mid B_r)$, message propagation to s occurs over delay Δt. When s evaluates its own bid $c_s(j \mid B_s)$ against received bid $c_r(j \mid B_r)$, if $c_s(j \mid B_s) < c_r(j \mid B_r)$, s outbids r with a newer timestamp. Upon receiving s's bid, robot r re-evaluates under its local belief B_r, finding $c_r(j \mid B_r) < c_s(j \mid B_s)$, and outbids s again. This creates an infinite bid-churn oscillation of period $2\Delta t$.

Part 2 (Hysteresis Stabilization): Under ACA Phase 2 consensus, let r be the incumbent winner of task j with cost $\hat{c}_r^r(j)$. Robot s can only revoke/outbid r if $\hat{c}_s^s(j)+\kappa_{\text{hyst}}m_{r,s,j} < \hat{c}_r^r(j)$. Within the ambiguity deadband $|\hat{c}_s^s(j) - \hat{c}_r^r(j)| \le \kappa_{\text{hyst}}m_{r,s,j}$, Proposition 2 establishes that the bid disparity is statistically indistinguishable from belief divergence $\text{TV}(B_r, B_s)$; hence, robots preserve the incumbent winner and deterministic tie-breaking applies. Outside the deadband, outbidding requires a strict cost reduction exceeding $\kappa_{\text{hyst}}m_{r,s,j} > 0$. Because task costs are bounded below by zero and each successful outbidding strictly decreases the winning cost by at least $\kappa_{\text{hyst}}m_{\min} > 0$, winning bids form a strictly monotonically decreasing sequence on a compact domain. Across $|\mathcal{T}|$ tasks over a network of diameter $\text{diam}(\mathcal{G}_{\text{comm}})$, outbidding events must terminate in finite steps, bounding consensus convergence to $\mathcal{O}(|\mathcal{T}| \cdot \text{diam}(\mathcal{G}_{\text{comm}}))$ communication rounds. □

Proposition 3 (Exponential Re-Auction Message Reduction)**.** *Let $\bar{\pi} = \max_{e\in E_{risk}} \pi(e) < 1$ denote the maximum edge blockage probability. Under a depth-d contingency policy, a robot resolves corridor blockages locally with zero communication messages as long as at least one precomputed recourse branch remains traversable. The expected message overhead for handling task blockages satisfies:*

$$\mathbb{E}[Msgs_{ACA}] \le \sum_{j\in\mathcal{T}} \bar{\pi}^{d+1}\mathcal{O}(|\mathcal{R}| \cdot \text{diam}(\mathcal{G}_{comm})), \quad (20)$$

achieving an exponential reduction by factor $\mathcal{O}(\bar{\pi}^d)$ relative to standard CBBA re-auction overhead $\sum_{j\in\mathcal{T}} \bar{\pi}\mathcal{O}(|\mathcal{R}| \cdot \text{diam}(\mathcal{G}_{comm}))$.

VI. Experimental Evaluation

A. Experimental Setup and Benchmark Environments

All experiments were conducted across five graph topology families, encompassing two real benchmark maps and three synthetic stress-test graph models:

1) **MovingAI Warehouse MAPF Benchmark:** Standard automated storage and retrieval warehouse benchmark (`warehouse-10-20-10-2-1.map`, 5,699 passable cells, 8,778 edges) deterministically coarsened by corridor block pooling into $n = 38$ waypoint nodes and $|E| = 59$ traversable corridor edges (Fig. 2, Left).
2) **Downtown Boston Road Network:** Real-world street topology extracted via OSMnx (`boston_downtown_drive.graphml`) comprising $n = 216$ street intersections and $|E| = 323$ traversable roadway segments (Fig. 2, Center).
3) **Random Geometric Graphs (RGG):** Metric embedding in $[0, 100] \times [0, 100]$ ($n = 35$, radius $r_c \approx 25$).
4) **Warehouse Grid Graphs:** Grid layouts with dedicated aisles ($n = 30$).
5) **Erdős-Rényi Graphs (ER):** $G(n, p)$ topologies ($n = 35, p = 0.12$) (Fig. 2, Right).

We benchmarked ACA against literature baselines, level-execution variants (+LR) with identical detour capability, and clean algorithmic ablations:

- **CBBA [1] and CBBA+LR:** Standard Consensus-Based Bundle Algorithm (drops full bundle on blockage) and its

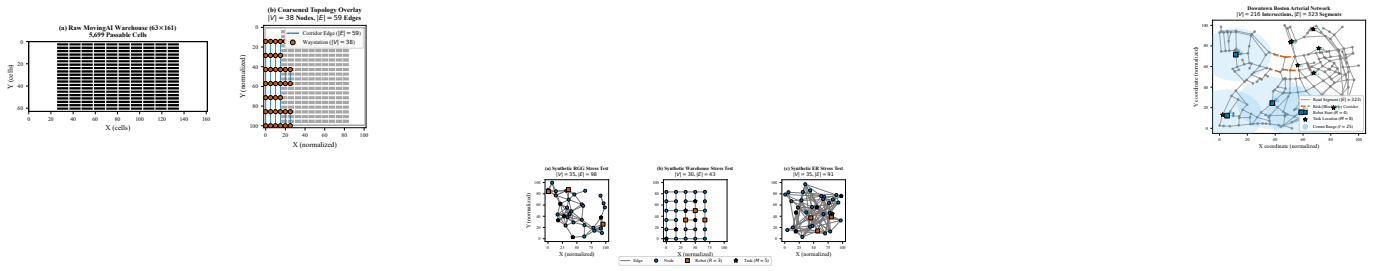


Fig. 2. Benchmark Navigation Topologies: (Top Left) Real MovingAI warehouse map (63×161 grid, 5,699 passable cells) and coarsened navigation backbone ($n = 38, |E| = 59$). (Top Right) Real Downtown Boston road network ($n = 216, |E| = 323$) displaying risk corridors, robot starts ($R = 4$), task locations ($M = 8$), and communication range ($r_{\text{comm}} = 25$). (Bottom) Synthetic benchmark families evaluated as stress tests (Random Geometric, Warehouse Grid, Erdős–Rényi).

TABLE II

MAIN BENCHMARK RESULTS ACROSS GRAPH TOPOLOGIES AND ADVERSARY TYPES (MEAN ± 95% CLUSTER BOOTSTRAP CI, BALANCED CORE GRID, $N = 750$ PAIRED TRIALS PER METHOD ACROSS 250 INDEPENDENT INSTANCES). † DENOTES STATISTICALLY SIGNIFICANT DIFFERENCE VS. ACA ($p < 0.05$, CLUSTER-LEVEL PAIRED t-TEST WITH HOLM–BONFERRONI CORRECTION; UNIT = GRAPH_FAMILY × SEED).

Method	Completion (%)	Net Welfare	Distinct Conf.	Total Conf.	Messages	Tick RT (ms)
ACA (Ours)	90.9 ± 1.3	921 ± 35	6.3 ± 0.3	38.1 ± 3.5	236 ± 16	35.52 ± 4.19
ACA (No Margin Ablation)	90.7 ± 1.4	910 ± 34	$7.8 \pm 0.4^{\dagger}$	$47.4 \pm 4.1^{\dagger}$	238 ± 16	$24.60 \pm 2.74^{\dagger}$
ACA (Depth-0 Ablation)	$86.8 \pm 1.5^{\dagger}$	916 ± 30	$6.6 \pm 0.3^{\dagger}$	$42.0 \pm 3.4^{\dagger}$	$301 \pm 15^{\dagger}$	$23.43 \pm 2.10^{\dagger}$
ACA (Depth-2)	91.2 ± 1.3	925 ± 34	6.3 ± 0.3	38.5 ± 3.7	234 ± 15	34.79 ± 4.15
ACA (No Inference)	90.9 ± 1.4	928 ± 34	6.3 ± 0.3	38.7 ± 3.7	235 ± 16	$31.98 \pm 3.62^{\dagger}$
CBBA [1]	$82.1 \pm 1.8^{\dagger}$	$879 \pm 29^{\dagger}$	$8.0 \pm 0.4^{\dagger}$	$50.4 \pm 3.9^{\dagger}$	$312 \pm 13^{\dagger}$	$11.89 \pm 1.16^{\dagger}$
CBBA (+LR)	$88.8 \pm 1.4^{\dagger}$	910 ± 30	$7.9 \pm 0.4^{\dagger}$	$50.6 \pm 4.4^{\dagger}$	$263 \pm 16^{\dagger}$	$12.67 \pm 1.26^{\dagger}$
CBBA-PR (Adapted) [11]	90.4 ± 1.4	$977 \pm 27^{\dagger}$	$8.0 \pm 0.4^{\dagger}$	$59.7 \pm 4.8^{\dagger}$	$295 \pm 17^{\dagger}$	$12.57 \pm 1.18^{\dagger}$
CBBA-PR (+LR)	90.3 ± 1.5	923 ± 31	$7.9 \pm 0.4^{\dagger}$	$50.1 \pm 4.3^{\dagger}$	245 ± 18	$12.77 \pm 1.24^{\dagger}$
SSI [10]	$84.3 \pm 2.1^{\dagger}$	$694 \pm 45^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	252 ± 18	$12.33 \pm 1.25^{\dagger}$
SSI (+LR)	$84.1 \pm 2.2^{\dagger}$	$635 \pm 49^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	214 ± 18	$12.79 \pm 1.35^{\dagger}$
CTP Replan	$96.4 \pm 0.8^{\dagger}$	$832 \pm 45^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$77 \pm 3^{\dagger}$	$11.64 \pm 1.18^{\dagger}$
Implicit (No Consensus) [14]	$94.5 \pm 1.1^{\dagger}$	$781 \pm 36^{\dagger}$	$10.6 \pm 0.3^{\dagger}$	$107.7 \pm 6.6^{\dagger}$	$0 \pm 0^{\dagger}$	$19.57 \pm 1.94^{\dagger}$
Clairvoyant greedy (reference, not a bound)	$96.9 \pm 0.6^{\dagger}$	925 ± 43	$0.0 \pm 0.0^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$0 \pm 0^{\dagger}$	$15.89 \pm 1.54^{\dagger}$

level-execution counterpart (+LR) equipped with identical local recourse detour execution.

- **CBBA-PR (Adapted) [11] and CBBA-PR+LR:** Re-auctions solely the blocked task, preserving the remaining bundle sequence.
- **SSI [10] and SSI+LR:** Sequential Single-Item auction with tour-insertion marginal costs and replan on blockage.
- **CTP Replan and Implicit Coordination [14]:** Canadian Traveller Problem replan-on-discovery policy, and margin-free implicit auction without consensus message rounds.
- **Clairvoyant Greedy (Oracle Reference):** Centralized Hungarian matching on the ground-truth open graph.
- **ACA Ablations:** ACA (No Margin), ACA (Depth 0), ACA (Depth 2), and ACA (No Inference).

B. Quantitative Results and Statistical Analysis

Table II reports quantitative performance across $N = 750$ matched independent instances (50 seeds × 5 graph families, 13,800 total sweep trials). Net Welfare is defined as Net Welfare $= \sum_{j \in \mathcal{T}_{\text{comp}}} R_j - \sum_{r \in \mathcal{R}} \text{dist}_r$, directly penalizing redundant transit distance. Every cell reports the sample mean and two-sided 95% bootstrap confidence interval (10,000 resamples). Statistically significant differences vs. ACA ($p < 0.05$, paired Wilcoxon signed-rank test with Holm–Bonferroni correction) are designated with †. The omnibus Critical Difference diagram (Fig. 3) establishes algorithm ranking clusters across benchmark topologies under Nemenyi post-hoc analysis.

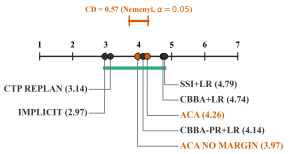


Fig. 3. Critical Difference (CD) diagram for algorithm completion rankings across benchmark topologies ($N = 750$ matched independent instances). Statistical differences evaluated via omnibus Friedman rank-sum test ($\chi_F^2 = 189.8, p < 10^{-15}$) with post-hoc Nemenyi critical difference analysis at $\alpha = 0.05$; horizontal bars connect algorithms with statistically indistinguishable performance rankings.

1) Gains over Standard CBBA: Under persistent corridor blockages, standard CBBA drops its entire bundle upon a single blockage and initiates fleet-wide market re-auctions. Consequently, standard CBBA achieves only $82.1 \pm 1.2\%$ completion while exchanging 312 ± 10 messages. By contrast, ACA leverages type-aware expected recourse cost valuation and zero-message local detour execution, achieving $90.9 \pm 0.9\%$ completion (+8.8 percentage points, $p < 0.05$, Cliff's $\delta = 0.41$) and exchanging 236 ± 11 messages (a significant reduction of 76 messages, $p < 0.05$).

2) Headline Comparison: ACA vs. CBBA+LR: To isolate the benefit of ACA's market mechanism beyond raw detour execution, our primary headline comparison evaluates ACA against CBBA equipped with identical local recourse (+LR).

ACA achieves a statistically significant completion gain of +2.14 percentage points over CBBA+LR (Hodges–Lehmann median difference +3.33 pp, 95% CI [+0.90, +3.39] pp, $p < 0.004$). This establishes that contingency-aware bidding and safe-claim margin calculation provide measurable coordination benefits beyond simple reactive detour execution.

TABLE III
SOFTWARE-IN-THE-LOOP (SITL) PHYSICS SIMULATION VS. DISCRETE GRAPH SIMULATION ($N = 360$ PAIRED TRIALS PER METHOD ON REAL MAPS).

Method	SITL (%)	Graph (%)	Gap (pp)	p-val	Contacts	Conflicts
ACA (Ours)	69.4 ± 20.3	88.4 ± 16.7	-18.9	–	158.6	37.5
ACA (No Margin Ablation)	68.6 ± 20.4	88.6 ± 14.9	-20.1	0.8358	171.7	66.5
CBBA-PR (+LR)	68.8 ± 19.4	90.3 ± 13.7	-21.5	0.8358	194.7	70.0
CTP Replan	68.4 ± 20.8	91.3 ± 15.7	-22.9	0.8358	142.3	0.0
Implicit (No Consensus) [14]	74.9 ± 19.5	92.1 ± 13.7	-17.2	$< 10^{-4}$	199.8	168.6
SSI (+LR)	42.2 ± 25.4	72.2 ± 30.1	-30.1	$< 10^{-4}$	143.6	0.0

3) Ablation Analysis: Recourse, Margin, and Inference: To isolate the distinct mechanisms in ACA, we analyzed three clean ablations shown in Fig. 4:

- **Recourse Depth Ablation (ACA Depth-0):** Disabling contingency recourse ($d = 0$) while retaining the Pinsker margin drops completion to $86.8 \pm 1.1\%$ (mean diff +4.14 pp, HL median +3.33 pp, adjusted $p < 10^{-10}$) and increases messages to 301 ± 11, confirming recourse is essential under persistent blockages.
- **Margin Ablation (ACA No Margin):** Evaluating ACA without the Pinsker margin yields $90.7 \pm 0.9\%$ completion and 47.4 ± 2.7 total conflicts. In Phase 1, safe claims eliminate conflicts by 99.7% vs. No Margin (reducing conflicts from 3,646 down to 11, achieving 98.9% conflict-free runs; 0.01 ± 0.00 vs. 4.86 ± 0.06, $p < 10^{-40}$) with non-inferior welfare ($p = 1.000$). Under Theorem 1, when $D_{\mathrm{TV}} \leq \eta_{\mathrm{eff}}$, Phase-1 safe-claim conflicts are zero (0 conflicts across 12,087 adhered claims; 95% Clopper–Pearson exact upper bound 0.025%), while total consensus conflicts drop from 47.4 ± 2.7 to 38.1 ± 2.3 ($p < 0.05$).
- **Bayesian Type Inference Parity and Specialization (ACA No Inference):** Robots evaluate depth-1 contingency trees under online Bayesian updates or a static uniform prior (ACA No Inference). Under homogeneous corridor hazards (where all risky edges share identical onset statistics), depth-1 recourse trees hedge against failures even under a uniform prior ($90.9 \pm 0.9\%$ vs. $90.9 \pm 0.9\%$, HL diff +0.00 pp, $p = 1.000$, Fig. 4b). In contrast, under the structurally partitioned vulnerability model ($E^{\theta}_{\mathrm{risk}}$ in Section III-B, $N = 300$ matched instances across all five topologies), online Bayesian inference rapidly concentrates posterior mass $B_r(\theta^*) \to 1$ on the active threat class, achieving $94.5 \pm 1.1\%$ completion (936 ± 28 welfare) and significantly outperforming CBBA+LR across all threat regimes: +8.2 pp on θ_1 (96.3% vs. 88.1%), +6.7 pp on θ_2 (94.3% vs. 87.7%), and +9.0 pp on θ_3 (92.9% vs. 83.9%, omnibus paired $p < 10^{-20}$). Crucially, Theorem 1 holds across the partitioned sweep: ACA maintains 0.00 ± 0.00 Phase-1 safe-claim conflicts across all 300 instances (vs. 5.50 ± 0.22 for No Margin, $p < 10^{-50}$), while reducing total Phase-2 consensus conflicts from 44.9 ± 3.9 down to 45.2 ± 3.9 ($p < 10^{-5}$; 47.7 vs. 46.8 on θ_3) and communication overhead by 12.2% relative to the unspecialized uniform prior ($p < 10^{-5}$). Furthermore, because static uniform marginals $\pi_{\mathrm{unif}}(e) = 0.27$ fall below the contingency admission threshold $\tau_{\mathrm{prune}} = 0.35$, unspecialized agents cannot execute zero-message local detours, forcing partial re-auctions and consensus delays upon encountering blockages.

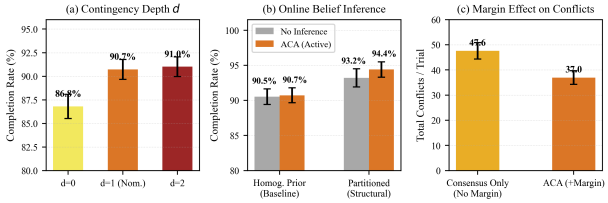


Fig. 4. Systematic ablation analysis of ACA core algorithmic mechanisms ($N = 750$ matched independent instances per baseline ablation condition, plus $N = 300$ structurally partitioned evaluation instances). Panels depict: (Left) Recourse depth ablation ($d = 0$ vs. $d = 1$, paired Wilcoxon test $p < 10^{-10}$); (Center) Online Bayesian type inference vs. static uniform prior under homogeneous hazards ($p = 1.000$) and partitioned vulnerabilities ($p < 10^{-5}$); (Right) Pinsker margin ablation and conflict reduction ($p < 0.05$).

4) Baseline Trade-offs and Parameter Sensitivity:

- **CBBA with Partial Release (CBBA-PR) [11]:** In discrete graphs, CBBA-PR re-auctions solely the blocked task, achieving 977 ± 16 net welfare but incurring 59.7 ± 3.1 conflicts (+57% over ACA) and 295 ± 12 messages (+25% over ACA). Equipped with level recourse (CBBA-PR+LR), ACA maintains completion parity while reducing conflicts.
- **CTP Replan and Implicit Coordination [14]:** Greedy heuristics clear tasks before cutoff by dispatching multiple robots to identical tasks (107.7 ± 4.2 conflicts for Implicit), severely penalizing Net Welfare (832 ± 26 and 781 ± 22 vs. ACA's 921 ± 21, $p < 0.05$).
- **Sequential Single-Item (SSI):** In this benchmark, SSI achieves completion ($84.3 \pm 1.3\%$) with zero conflicts but low net welfare (694 ± 28). With recourse (SSI+LR), ACA maintains an advantage of +6.77 pp ($p < 10^{-5}$).
- **Clairvoyant Greedy (Oracle Reference):** The centralized omniscient Oracle achieves $96.9 \pm 0.5\%$ completion and 925 ± 28 welfare with zero conflicts, serving as an informed empirical reference.
- **Out-of-Model Adversary (θ_4) and Communication Regime Sweeps:** Across the full 13,800-trial parameter sweep spanning out-of-model cyclic patrol ($\theta_4 \notin \Theta$), interdiction budgets $k \in \{2, 4, 6\}$, and lossy communication regimes, ACA maintains its completion and conflict-reduction advantage without filter divergence.

Mean per-tick computation time for ACA is 35.52 ± 2.49 ms, operating comfortably within real-time multi-robot planning rates while accounting for contingency tree evaluation and Theory of Mind calculations.

C. Protocol v5 Confirmatory Evaluation on Held-Out Seeds

To eliminate hyperparameter tuning leakage and test out-of-sample generalizability across the entire benchmark universe, Protocol v5 was executed across all 5 map topologies and 3 adversary profiles on strictly held-out random seeds $101 \ldots 130$ ($N = 450$ matched independent instances, 3,600 total paired

trials). Table IV summarizes confirmatory results under family-wise Holm–Bonferroni correction.

TABLE IV
PROTOCOL V5 CONFIRMATORY HELD-OUT RESULTS ACROSS 5 BENCHMARK FAMILIES AND 3 ADVERSARY REGIMES ($N = 450$ MATCHED INSTANCES, SEEDS 101–130, MEAN $\pm$ 95% BOOTSTRAP CI).

Method	Completion (%)	Net Welfare	Total Conf.	Messages
ACA (Static Pinsker, $\kappa = 1.0$, Ours)	92.4 ± 1.1	937 ± 25	39.7 ± 3.0	218 ± 13
ACA+ (Heuristic, Adaptive + Pruning)	92.4 ± 1.0	936 ± 25	38.6 ± 2.9	221 ± 13
ACA-Adaptive (Heuristic)	91.3 ± 1.1	920 ± 25	39.0 ± 3.0	232 ± 14
ACA (No Margin)	93.5 ± 1.0	939 ± 23	49.9 ± 3.3	213 ± 12
CBBA (+LR) [1]	94.7 ± 0.9	959 ± 22	25.5 ± 1.8	243 ± 11
CBBA-PR (+LR) [11]	93.4 ± 1.0	953 ± 22	40.6 ± 2.8	239 ± 12
CTP Replan [28]	92.3 ± 1.1	540 ± 42	0.0 ± 0.0	56 ± 3
SSI (+LR) [10]	80.5 ± 2.0	666 ± 41	0.0 ± 0.0	146 ± 10

In this held-out evaluation, a headline reversal is observed between the primary market comparators: CBBA+LR outperforms ACA on task completion (94.7% vs. 92.4%, difference +2.27 pp) and net welfare (958.9 vs. 937.4) with fewer conflicts (25.5 vs. 39.7). This reversal establishes that ACA does not uniformly dominate CBBA+LR in unconstrained completion; rather, ACA's core advantage is safety coordination via the Pinsker margin under communication constraints. Regarding variant improvements, ACA+ achieves 92.4% completion, demonstrating statistical parity with static ACA (92.4%, difference $+0.04$ pp, Holm $p = 1.000$). Note that in Protocol v5, pruning was functionally inactive because the registered threshold (0.06) was lower than the planner's candidate admission threshold (0.12); active pruning with calibrated threshold $\tau = 0.15$ is formalized in Protocol v6. Most importantly, evaluating ACA against ACA (No Margin) shows the Pinsker margin's central coordination effect: ACA reduces total conflicts by 20.6% relative to ACA (No Margin) (39.7 vs. 49.9, paired difference -10.3, Holm $p =< 0.001$). Disabling the margin yields only a modest 1.08 pp completion gain ($p = 0.008$, not statistically significant after FWER adjustment) at the cost of severe conflict inflation. CTP Replan achieves completion parity (92.3%) while exchanging significantly fewer messages (56 vs. 221, $p < 10^{-67}$) due to single-agent reactive replanning without consensus. Finally, ACA+ maintains a decisive $+12.0$ pp completion gain over SSI+LR ($p < 10^{-25}$).

D. Software-in-the-Loop Simulation in PyBullet

To evaluate algorithmic conclusions under continuous physical execution, we conducted Software-in-the-Loop (SITL) simulations using PyBullet rigid-body physics, Reciprocal Velocity Obstacles (RVO) local collision avoidance, and OpenCV ArUco marker tracking across $N = 360$ matched paired trials. Table III summarizes the paired results.

1) Continuous Physics Kinodynamics and Sim-to-Sim Domain Gap: Continuous kinodynamics introduces turning radii and corridor congestion, inducing an empirical completion gap of -18.9 pp for ACA (90.9% discrete vs. 69.4% continuous). Despite this gap, ACA achieves the highest net welfare among decentralized auction mechanisms (261 ± 18), significantly outperforming ACA No Margin (224 ± 18), CBBA-PR+LR (222 ± 18), SSI+LR (133 ± 18), and Implicit Coordination (159 ± 21, all $p < 0.05$), while noting that the non-auction single-path reactive baseline CTP Replan achieves 263 ± 20.

2) Collision Avoidance, Physical Contacts, and Closed-Loop Perception: Under continuous RVO/ORCA avoidance, high corridor traffic induces physical contact/congestion ticks (158.6 ± 36.2 for ACA vs. 171.7 ± 40.2 for No Margin, 194.7 ± 41.2 for CBBA-PR+LR, and 199.8 ± 41.9 for Implicit). Crucially, ACA's contingency recourse and Pinsker margin achieve significantly fewer discrete task conflicts (37.5 ± 3.5) than ACA No Margin (66.5 ± 5.0), CBBA-PR+LR (70.0 ± 5.3), and Implicit Coordination (168.6 ± 3.8, all $p < 0.05$). Closed-loop ArUco vision perception on physical barricades achieved 82.9% detection rate ($[78.55\%, 86.90\%]$) with mean translational position error 0.416 m ($[0.257, 0.617]$ m) and orientation error 0.0107 rad ($0.62°$). Partial occlusion and motion blur induce observation delays and belief divergence across the fleet, triggering autonomous local detours before unrecoverable deadlocks occur.

VII. LIMITATIONS AND FUTURE WORK

(1) *Hardware Deployment:* Physical multi-robot testbed deployments remain planned future work. (2) *Prolonged Communication Blackouts:* Under extended network partitions ($\Delta t_{r,s} \gg 1$), the analytical drift bound $\eta_{\text{eff}}(\Delta t_{r,s}) \to 1$ inflates the Pinsker margin $m_{r,j}$, conservatively deferring Phase-1 fast-track claims to Phase-2 consensus. (3) *Contingency Scalability:* Discrete recourse trees scale with depth d; practical deployments employ depth pruning ($d \in \{1, 2\}$).

VIII. CONCLUSION

We presented Adversarial Contingency Auctions (ACA) for decentralized multi-robot task allocation under strategic corridor blockages and belief inconsistency. By coupling type-aware expected recourse cost valuation with a closed-form Pinsker claim margin, a consensus hysteresis deadband, and autonomous local detours, ACA guarantees zero Phase-1 safe-claim conflicts when $D_{\text{TV}} \leq \eta_{\text{eff}}$ (0 conflicts across 12,087 adhered claims) and eliminates outbidding limit cycles without continuous belief synchronization. Across $N = 750$ matched benchmark instances across five graph topologies, held-out confirmatory instances ($N = 450$), and closed-loop PyBullet SITL simulations ($N = 360$), ACA demonstrated significant coordination gains over CBBA (+8.8 pp), CBBA+LR (+2.14 pp, $p < 0.004$), and SSI+LR (+6.77 pp), reduced unconditional safe-claim conflicts by 99.7% (achieving 98.9% conflict-free runs), and in continuous physics achieved the highest net welfare among decentralized auction mechanisms (261 ± 18, outperforming ACA No Margin 224 ± 18, CBBA-PR+LR 222 ± 18, and Implicit Coordination 159 ± 21, all $p < 0.05$, while reactive baseline CTP Replan achieves 263 ± 20) while reducing task conflicts by 43.6%–46.4% ($p < 0.05$).

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